Reflective projective billiards in the plane: odd periods, period five, and metric rigidity

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Abstract

We study polygonal projective billiards whose reflection law is constant on each side. We give a single explicit family with an open set of periodic trajectories for every period at least three, including every odd period, and identify the previously known centrally projective examples as a special parameter value. For period five we describe the full scalar-monodromy variety, prove its irreducibility, and show that every configuration preserves a nonzero quadratic form. Under the stated distinct-centre and non-collinearity hypotheses that form is nondegenerate; this yields an explicit unirational classification of the convex five-reflective locus. We also relate the definite members to spherical billiards and prove that, among analytic simply connected constant-curvature surfaces with geodesic walls, only the sphere can carry an open family of periodic trajectories. For general complete analytic ambients with global wall reflections we prove compactness and a common period for the geodesic flow. The arguments reduce the billiard return map to products of harmonic homologies and separate the algebraic closure condition from the strict first-impact inequalities.

Keywords. Projective billiards; periodic trajectories; harmonic homologies; character varieties; spherical billiards.

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1 Introduction

Ivrii's conjecture [7] predicts that periodic trajectories in an ordinary Euclidean billiard have measure zero. Projective billiards replace perpendicular reflection by a harmonic projective involution and admit reflective tables, that is, tables with an open set of trajectories of one fixed period. Fierobe constructed the right-spherical triangle and examples of every even period at least four [4, 5]. Problem 1 of [2] asks whether odd periods at least five occur and whether further examples exist for a fixed period.

The first main result answers the existence question for every period. The second main result gives a complete algebraic and convex classification at period five. The proofs use the matrix model $H = I - 2OL/(LO)$ for a harmonic homology. Scalar monodromy supplies closure for all nearby rays, while strict inequalities for one witness orbit turn that algebraic relation into a genuine open set of billiard trajectories.

The final part explains the metric meaning of the construction. A definite invariant conic gives a spherical billiard in projective coordinates. Conversely, a complete analytic Riemannian ambient carrying such an open family, with global isometric wall reflections, has

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periodic geodesic flow and is compact. This also resolves negatively the apparent possibility of a reflective billiard contained in the Klein disc.

For classical background on manifolds all of whose geodesics are closed, see [6, 1].

The main contributions of this paper are: an explicit convex family with a primitive open family for every period $n \geq 3$, including all odd periods; a complete algebraic and convex classification of the five-reflective locus, including its invariant conic and irreducibility; and a metric rigidity theorem giving global periodicity of the geodesic flow, together with spherical rigidity among simply connected constant-curvature surfaces. The distinction between structural theorems, exact rational certificates, and the previously known centrally projective examples is maintained throughout.

All symbolic certificates mentioned below concern explicit rational data only; the structural results and classification proofs are analytic or algebraic.

2 Setting

A *projective billiard* [8] is a bounded domain $\Omega \subset \mathbb{R}^2$ together with a *transversal line field* assigning to each $p \in \partial\Omega$ a line $L(p)$ through p not tangent to $\partial\Omega$; reflection is defined by requiring the incoming ray, the outgoing ray, the tangent and $L(p)$ to have cross ratio -1.

Throughout this paper we restrict to *central facet fields*: on each facet the transversal lines belong to the pencil through one fixed projective centre outside the supporting hyperplane. No reduction from arbitrary transversal fields to this class is asserted. Statements about the period spectrum use the stated convention on the facets visited by the witness.

Definition 2.1 (Fierobe)**.** Ω is *k-reflective* if there are a k-periodic orbit $(p_1, \dots, p_k)$ and an **open** $U_1 \times U_2 \subset (\partial\Omega)^2$ containing (p_1, p_2) every point of which begins a k-periodic orbit.

The open set is the whole content of the definition, and it is where the difficulty lives: exhibiting one periodic orbit is easy, exhibiting a two-parameter family of them is not.

2.1 Reflection as a harmonic homology

For a covector L and a vector O with $LO \neq 0$ put $H = I - 2\,OL/(LO)$. Then $H^2 = I$, $HO = -O$, $Hx = x$ for $Lx = 0$, and $\det H = -1$.

Proposition 2.2. *Let p lie on L, let τ span L, $\nu = O - p$. The involution induced by H on the pencil at p sends $a\tau + b\nu$ to $a\tau - b\nu$; hence the cross ratio of (incoming, outgoing; tangent, transversal) is -1. So reflection in the side with transversal field the pencil through O* is *the harmonic homology with axis the side and centre O.*

Proof. H fixes p and is an involution, so induces an involution of the pencil at p fixing the two distinct lines L and pO; in the coordinate taking them to 0 and ∞ it is $m \mapsto -m$. Parametrising by $(a : b)$ with direction $a\tau + b\nu$, the tangent is $(1 : 0)$, the transversal $(0 : 1)$, and with $[ij] = a_i b_j - a_j b_i$ the cross ratio of $(a : b), (a : -b), (1 : 0), (0 : 1)$ is $[13][24]/([14][23]) = -1$. □

Consequently, if $H_{j_k} \cdots H_{j_1}$ is scalar then *every* line returns to itself after the corresponding k reflections. The converse is false: an isolated trajectory may well close without the product being scalar, and it is precisely the scalar case that produces an *open set* of closed trajectories, which is what Definition 2.1 demands. As $\det \prod H = (-1)^k$, for odd k the relevant value is $-I$, which is the identity of PGL_3: there is no determinant obstruction.

For the central fields considered here, an open family also forces scalar monodromy. The initial lines of a regular family form an open subset of the Grassmannian of projective

lines. Their invariance under the fixed product is an algebraic incidence condition; vanishing on an open subset forces it on the entire Grassmannian. A linear map preserving every two-dimensional vector subspace preserves every one-dimensional subspace, obtained as an intersection of such subspaces in vector dimension at least three. Applying this to basis vectors and their pairwise sums shows that the map is scalar. This necessity concerns an open family; an isolated closed orbit does not suffice.

3 An openness criterion

Everything below rests on this; it assumes no symmetry.

Lemma 3.1 (Exit map)**.** *Let Ω be a convex polygon, p interior to a side, w pointing into Ω. The ray $p+sw$, $s>0$, meets $\partial\Omega$ in exactly one point $E(p,w)$, and E is continuous wherever $E(p,w)$ is not a vertex.*

Proof. Ω is convex and bounded, so $s^* = \sup\{s>0 : p+sw\in\Omega\}$ is finite and $E(p,w) = p+s^*w \in \partial\Omega$; uniqueness is convexity. If $E(p,w)$ is interior to a side a then w is not parallel to its supporting line, so the intersection determinant is nonzero and Cramer's rule gives s^* near (p,w) as a quotient of polynomials with nonvanishing denominator. Since the interior of a is open in $\partial\Omega$, nearby data still exit through a. □

Remark 3.2. Requiring nonzero intersection determinants with every supporting hyperplane is unnecessarily restrictive: a strict trajectory may be parallel to facets it never reaches. Continuity only requires transversality at the actual exit facet, together with persistence of the strict inequalities for the other facets.

Theorem 3.3 (Openness criterion)**.** *Let Ω be a convex polygon with sides $a_0,\dots,a_{k-1}$ and transversal field on a_i the pencil through O_i, with $L_iO_i \neq 0$. Assume* (H1) *$H_{k-1}\cdots H_0 = cI$, $c\neq 0$;* (H2) *there is an orbit with itinerary $0,1,\dots,k-1$ whose impacts are strictly interior to their sides, with strictly positive step parameters, each losing side excluded by a strict inequality and nonvanishing transversality at each impact;* (H3) *the return direction is $u_k = \kappa u_0$ with $\kappa>0$. Then Ω is k-reflective; if the k impacts lie on distinct sides the period is primitive.*

Proof. Each step is a line–line intersection followed by a reflection, so the impact parameters and directions are rational functions of the initial data, with denominators nonvanishing at the witness by (H2); the vanishing locus being closed, they are defined and continuous nearby. The finitely many strict inequalities of (H2) persist, and by Lemma 3.1 each of the k exit maps is continuous with value in the interior of a side, which is open in $\partial\Omega$; composing, the itinerary is unchanged for nearby data. By (H1) the product is the identity of PGL_3, so *every* line returns to itself: $\ell_k = \ell_0$ as an unoriented line, and since ℓ_0 is not the supporting line of a_0 the intersection $\ell_0 \cap a_0$ is a single point, whence $p_k = p_0$. For the orientation, (H1) only gives $u_k = \kappa u_0$ with κ real nonzero — neither ±1 nor a priori locally constant — but $\kappa = (u_k\cdot u_0)/(u_0\cdot u_0)$ is continuous and nonvanishing and equals a positive number at the witness by (H3), so $\kappa>0$ on a smaller neighbourhood. Finally, a proper period $d \mid k$, $0<d<k$, would force $p_d = p_0$; but p_d lies on side a_d and p_0 on a_0, which are distinct sides, and the impacts are interior to them, so $p_d \neq p_0$. □

Hypotheses (H2)–(H3) are finitely many strict inequalities and (H1) one identity, so they can be certified exactly for explicit data. We use this twice.

4 One family for every n

Fix $n \geq 3$, put $\alpha = \pi/n$, $u = \sin\alpha$, $v = \cos\alpha$, and let the regular n-gon have vertices $V_j = (\cos 2j\alpha, \sin 2j\alpha, 1)^\top$, side i being V_iV_{i+1} with supporting covector L_i and midpoint M_i. Let $\widehat{M_i}$ be the unit vector from the centre towards M_i and set

$$O_i = -\lambda\,\widehat{M_i}, \tag{1}$$

the transversal field on side i being the pencil through O_i. No parity is involved: O_i is simply a point of the line joining the centre to M_i. Note that λ parametrises the *ansatz*, not a continuum of n-reflective billiards: for each n only the finitely many values $\lambda(n, j)$ of Theorem 4.6 give n-periodicity.

Remark 4.1. For n odd, $\widehat{M_i}$ is antipodal to the vertex direction V_{i+m} with $m = (n+1)/2$ — the unique solution of $2m \equiv 1 \pmod n$ — so (1) reads $O_i = \lambda V_{i+m}$, "the opposite vertex pushed out by λ". For $\lambda = 0$ all O_i are the centre, which is Fierobe's centrally projective construction, defined for every n. Formulation (1) covers both.

From $V_0 \times V_1$, using $\sin 2\alpha = 2uv$ and $\cos 2\alpha - 1 = -2u^2$, one gets $L_0 \propto (-v, -u, v)$, while $\widehat{M_0} = (v, u)$ gives $O_0 = (-\lambda v, -\lambda u, 1)^\top$ and

$$L_0O_0 = \lambda + v. \tag{2}$$

Let R be the rotation by 2α, so $R^n = I$ and $RO_i = O_{i+1}$.

Lemma 4.2. *$H_i = R^iH_0R^{-i}$, and with $W = H_0R^{-1}$ one has $P := H_{n-1}\cdots H_0 = R^{-1}W^nR$.*

Proof. The homology with axis $g(L)$ and centre $g(O)$ is gHg^{-1}; take $g = R^i$. Induction gives $H_{k-1}\cdots H_0 = R^{k-1}(H_0R^{-1})^{k-1}H_0$, since $H_k\big(R^{k-1}(H_0R^{-1})^{k-1}H_0\big) = R^k(H_0R^{-1})^kH_0$. Take $k = n$, use $R^n = I$ and $H_0 = WR$. □

Lemma 4.3 (Polarity)**.** *If $L = \mu\,O^\top Q$ for a nondegenerate Q and $\mu \neq 0$ (i.e. L is the polar of O), then $H \in \mathrm{O}(Q)$.*

Proof. μ cancels: $H = I - 2OO^\top Q/s$ with $s = O^\top QO$, so $H^\top = I - 2QOO^\top/s$ and $H^\top QH = Q - 4QOO^\top Q/s + 4QO\,s\,O^\top Q/s^2 = Q$. □

Lemma 4.4. *With $c_2 = \cos 2\alpha$:* (a) $\det W = -1$*;* (b) $\operatorname{tr} W = (\lambda + \cos 3\alpha)/(\lambda + \cos\alpha)$*;* (c) *$W$ preserves $Q = \operatorname{diag}(1, 1, \lambda v)$;* (d) $\chi_W(t) = -(t+1)(t^2 - (\operatorname{tr} W + 1)t + 1)$.

Proof. (a) $\det W = \det H_0 \det R^{-1} = (-1)(1)$; no computation of W is needed.

(b) $W = R^{-1} - \frac{2}{D}O_0L_0R^{-1}$ with $D = \lambda + v$, so $\operatorname{tr} W = \operatorname{tr} R^{-1} - \frac{2}{D}\operatorname{tr}(O_0L_0R^{-1})$, and for a rank-one matrix $\operatorname{tr}(O\,L\,M) = L\,M\,O$. With $L_0 = (-v, -u, v)$ and R^{-1} the rotation by -2α,

$$L_0R^{-1} = \big(v(4u^2-1),\ -u(4v^2-1),\ v\big), \qquad L_0R^{-1}O_0 = \lambda(v^2-u^2) + v = \lambda c_2 + v,$$

the terms in $4\lambda u^2v^2$ cancelling. Since $\operatorname{tr} R^{-1} = 2c_2 + 1$,

$$\operatorname{tr} W = (2c_2+1) - \frac{2(\lambda c_2 + v)}{\lambda + v} = \frac{\lambda + 2vc_2 - v}{\lambda + v},$$

and $2vc_2 - v = v(4v^2 - 3) = \cos 3\alpha$.

(c) By Lemma 4.3 it suffices that L_0 be the polar of O_0, which is one line: $QO_0 = (-\lambda v, -\lambda u, \lambda v)^\top = \lambda L_0^\top$; and R preserves Q, being a rotation in (x, y) fixing z.

(d) Suppose first $\lambda \neq 0$, so that Q is nondegenerate. Then $W \in \mathrm{O}(Q)$ gives $W^{-1} = Q^{-1}W^{\top}Q$, similar to $W^{\top}$, so the spectrum is invariant under $\mu \mapsto \mu^{-1}$; in dimension 3 some eigenvalue is fixed, hence ± 1, and $\det W = -1$ forces it to be -1. The other two are roots of $t^2 - \beta t + 1$ with $\beta = \operatorname{tr} W + 1$ by the trace.

For $\lambda = 0$ the form $Q = \operatorname{diag}(1, 1, 0)$ is degenerate and this argument does not apply. But (d) is an identity between polynomials in λ, u, v modulo $u^2 + v^2 - 1$ — both sides are polynomial in the entries of W, which are rational in λ with denominator $\lambda + v$, nonvanishing at $\lambda = 0$ — and a polynomial identity valid on the Zariski-dense set $\lambda \neq 0$ holds identically. Hence (d) holds for every λ, including the value $\lambda = 0$ of Proposition 4.8. □

Remark 4.5. The case $\lambda = 0$ is worth isolating because it is exactly Fierobe's centrally projective family (Proposition 4.8): there all O_i are the centre of the polygon, the invariant "conic" degenerates to the doubled line at infinity, and the underlying geometry is Euclidean rather than spherical or hyperbolic.

Theorem 4.6. *Let $n \geq 3$ and let j be an integer with $1 \leq j < n/2$. Put $\theta = (n - 2j)\pi/n \in (0, \pi)$, $C = \cos\theta$ and*

$$\lambda = \lambda(n, j) = \frac{v\,(2v^2 - 1 - C)}{C - 1}. \tag{3}$$

Then the projective billiard (1) *on the regular n-gon is n-reflective, with primitive period n.*

Proof. Algebraic closure. First $\lambda + v = v(2v^2 - 2)/(C - 1) = 2vu^2/(1 - C) > 0$ since $C \in (-1, 1)$, $v > 0$, $u \neq 0$; so by (2) the transversal field is admissible. Substituting (3) in Lemma 4.4(b) gives $(\operatorname{tr} W + 1)/2 = C$, so by Lemma 4.4(d) the eigenvalues of W are -1 and the roots of $t^2 - 2Ct + 1$, namely $e^{\pm i\theta}$; they are distinct from each other and from -1, so W is diagonalisable and, as $n\theta = (n - 2j)\pi \equiv n\pi \pmod{2\pi}$,

$$W^n = \operatorname{diag}\big((-1)^n, e^{in\theta}, e^{-in\theta}\big) = (-1)^n I,$$

which is the identity of PGL_3 for either parity. Hence $P = (-1)^n I$ by Lemma 4.2.

The midpoint orbit. By (1) the points O_i, the centre and M_i are collinear, and in a regular polygon the radius to the midpoint of a side is perpendicular to it; so the transversal at M_i is normal to side i, and there the projective law is the optical one by Proposition 2.2. The midpoint polygon is the classical periodic orbit of the Euclidean billiard in the regular n-gon — the perpendicular bisector of side i is a symmetry axis fixing M_i and swapping M_{i-1}, M_{i+1} — and traversing it once returns the same direction, so $\kappa = 1$.

Admissibility. All impacts are at $t = 1/2$, strictly interior, so none is a vertex. A polygon is not strictly convex, so we argue with convexity alone: M_i and M_{i+1} are interior to distinct sides, so the chord $[M_i, M_{i+1}]$ is not contained in $\partial\Omega$; as Ω is convex with both endpoints on $\partial\Omega$, its interior lies in the interior of Ω, for otherwise convexity would put the whole segment in the boundary. Hence M_{i+1} is the first boundary point met, every other side being excluded strictly.

Theorem 3.3 now applies, the n impacts lying on n distinct sides. □

Corollary 4.7. *There are k-reflective projective billiards for every $k \geq 3$, of either parity; in particular for every odd $k \geq 5$.*

Proposition 4.8 (Where Fierobe's family sits)**.** *$\lambda(n, j) = 0$ if and only if $\theta = 2\alpha$, i.e. $j = (n - 2)/2$, which is an integer if and only if n is even. Thus the centrally projective construction is the $\lambda = 0$ member of the family exactly for even n, and for odd n it does not belong to the family.*

Proof. $\lambda = 0$ forces $2v^2 - 1 = C$, i.e. $\cos 2\alpha = \cos\theta$ with both angles in $(0, \pi)$, hence $\theta = 2\alpha$ and $(n - 2j)/n = 2/n$. □

Remark 4.9. This is exactly [4, Prop. 1.5]: for odd n the centre is not an n-reflective member, and the construction yields period $2n$ instead. The invariant conic is $x^2+y^2 = -\lambda\cos(\pi/n)z^2$: these are billiards of constant curvature in projective coordinates, and which curvature is decided by the sign of λ, as follows.

Proposition 4.10 (Which geometry, and where the mirrors sit)**.** *Since $C < 1$, the sign of $\lambda(n, j)$ is that of $C - \cos 2\alpha$, so*

$$\lambda > 0 \iff \theta < 2\alpha \iff j > \tfrac{n-2}{2}.$$

With $1 \le j < n/2$ an integer this gives: for n odd, exactly one member with $\lambda > 0$, namely $j = (n - 1)/2$; for n even, none, and $j = (n - 2)/2$ is the Euclidean $\lambda = 0$ member of Proposition 4.8; every other member has $\lambda < 0$. Moreover, whenever $\lambda < 0$ the invariant conic is a real circle of radius$^2 = -\lambda\cos\alpha$, and

$$-\lambda < \cos\alpha \iff \frac{\cos 2\alpha - C}{1 - C} < 1 \iff \cos 2\alpha < 1,$$

*which always holds; so radius$^2 < \cos^2\alpha$, the conic lies strictly inside the inscribed circle of the polygon, and **no mirror meets it**. Sharper, with $u = \sin\alpha$, $v = \cos\alpha$:*

$$-\lambda < v^3 \iff \cos 2\alpha - v^2 < C\,(1 - v^2) \iff -u^2 < C\,u^2 \iff C > -1,$$

*also always true; so radius$^2 < v^4$, which is the inradius^2 of the midpoint orbit, and the conic lies strictly inside the region enclosed by **every orbit of the family**.*

Remark 4.11 (The three geometries, stated carefully). For $\lambda > 0$ the form is definite and the geometry is genuinely spherical, and $\lambda = 0$ is the Euclidean case. For $\lambda < 0$ the form has signature $(2, 1)$, the group is $\mathrm{O}(2, 1)$, and the absolute is a real conic — and here one must be careful, because the domain does *not* sit inside one Cayley–Klein component. By Proposition 4.10 the conic lies strictly inside the domain, so the domain *straddles* the absolute: part of it is the Klein disc, where the invariant metric is Riemannian of curvature -1, and part is the exterior, where the metric is Lorentzian. No single metric makes the whole domain a metric billiard, and the invariant metric degenerates along the conic, which is an interior curve of the domain. So calling the $\lambda < 0$ branch "hyperbolic" would be wrong; what is true, and all that is used anywhere in these notes, is that a nondegenerate conic is preserved.

The orbits, on the other hand, do sit in one component: by the sharper inequality of Proposition 4.10 the conic lies strictly inside the hole enclosed by every orbit of the family, so no orbit ever crosses the absolute and the dynamics of the family is de Sitter even though the domain is not.

There is in fact no 5-reflective projective billiard, among those satisfying the hypotheses of Theorem 5.17 — convex pentagon, pairwise distinct centres, no four centres collinear — whose domain lies *inside* the Klein disc of its invariant conic. Indeed, by that theorem such a configuration preserves a nondegenerate conic and each centre is the pole of its mirror, so a domain inside the Klein disc would be a hyperbolic billiard with totally geodesic walls, and Theorem 5.26 below excludes k-reflectivity for every k. Thus the indefinite members above necessarily straddle the absolute; none is a genuine hyperbolic-plane billiard.

Remark 4.12. For $n = 5$, $j = 2$: $\lambda = (2 + \sqrt{5})/2$. The $\sqrt{5}$ is forced by the pentagon, not by the construction: $\lambda(n, j)$ is a rational function of $\cos(\pi/n)$ and so lies in $\mathbb{Q}(\cos(\pi/n))$, which for $n = 5$ is $\mathbb{Q}(\sqrt{5})$ and for $n = 7, 9$ is a cubic field.

5 Five reflections: examples, moduli and classification

5.1 Imposing a conic makes the problem rational

By Lemma 4.3, if all k homologies preserve a conic Q then each centre is the pole of its axis, so a single vector n_i determines the mirror; with $Q = x^2 + y^2 + z^2$ the homology is the ordinary reflection $R_i = I - 2n_i n_i^\top/(n_i \cdot n_i)$, rational whenever n_i is, and the closure condition becomes a product of k reflections in $\mathrm{O}(3)$ equal to $\pm I$. For $k = 5$ this can be solved rather than searched.

Proposition 5.1. *Let $n_0, n_1, n_2 \in \mathbb{Q}^3$ with $B := -R_2R_1R_0 \neq I$, and let w span $\ker(B - I)$. Choose $n_4 \in \mathbb{Q}^3$ with $n_4 \cdot w = 0$. Then $R_3 := BR_4$ is a reflection with rational normal, and $R_4R_3R_2R_1R_0 = -I$.*

Proof. $\det R_i = -1$, so $\det B = +1$ and B is a rotation; $w \in \ker(B - I)$ is rational by Gaussian elimination. From $n_4 \cdot w = 0$ we get $R_4 w = w$, and a reflection whose mirror plane contains the axis conjugates the rotation to its inverse: $R_4BR_4 = B^{-1}$. Hence $(BR_4)^2 = B(R_4BR_4) = I$ and $\det(BR_4) = -1$. An involution of $\mathrm{O}(3)$ of determinant -1 is either a reflection or $-I$, and $BR_4 = -I$ would give $B = -R_4$, whence $\ker(B-I) = \ker(R_4+I) = \langle n_4 \rangle$ and $n_4 \cdot w = 0$ would read $n_4 \cdot n_4 = 0$; so $R_3 := BR_4$ is a reflection, its normal spans $\ker(R_3+I)$ and is rational. Finally $R_3R_4 = B = -R_2R_1R_0$ gives the stated identity. □

5.2 A second, inequivalent 5-reflective billiard

Theorem 5.2. *The configuration $n_0 = (1, 0, -3)$, $n_1 = (3, 2, 2)$, $n_2 = (-1, 0, 1)$, $n_3 = (-8, 9, 5)$, $n_4 = (0, -2, 2)$ defines a 5-reflective projective billiard, not projectively equivalent to the $n = 5$ member of Theorem 4.6.*

Proof. The hypotheses of Theorem 3.3 are verified in exact rational arithmetic: (H1) the product of the five reflections is $-I$ over $\mathbb{Q}$, computed in both orders; the five successive cross products of consecutive edge vectors share a nonzero sign, so the pentagon is convex with sides in cyclic order and no vertex at infinity; (H2) the witness orbit with $t_0 = t_1 = 1/14$ has at each step a first hit on the next side with $0 < t < 1$ and $s > 0$ strictly, every losing side excluded strictly, and nonvanishing transversality; (H3) $u_5 = \kappa u_0$ with $\kappa > 0$, and $t_5 - t_0 = 0$ exactly. The five impacts lie on five distinct sides.

For inequivalence, the $n = 5$ member of Theorem 4.6 has a projective symmetry of order 5 cycling its mirrors, which is a projective invariant; determining the unique $g \in \mathrm{PGL}_3$ with $n_0 \mapsto n_1 \mapsto \cdots \mapsto n_4$ and checking $g(n_4) \not\propto n_0$ shows the present configuration has none. □

5.3 The case $k = 4$, and its failure for $k = 6$

Theorem 5.3. *For $k = 4$, every configuration whose homologies preserve a common conic is degenerate: the normals are coplanar, hence the four lines concurrent.*

Proof. $R_0R_1R_2R_3 = I$ gives $R_0R_1 = R_3R_2$, each side a product of *exactly two* reflections, hence a rotation with axis $n_0 \times n_1$ resp. $n_3 \times n_2$. Equal rotations have equal axes, so all four n_i are orthogonal to a common vector. □

Remark 5.4. The same statement for general even k is false. For $k = 6$ one gets $R_0R_1R_2 = R_5R_4R_3$, three reflections on each side, a factorisation that always exists. Explicitly $n = (-2, 3, 3), (-3, -1, -1), (-2, -2, 3), (1, -3, 2), (2, 1, -3), (3, 4, 3)$ satisfies $R_0 \cdots R_5 = +I$ exactly over $\mathbb{Q}$, preserves $x^2 + y^2 + z^2$, and has normals of rank 3.

5.4 Dimensions

Let V_k be the variety of configurations of k lines with k centres whose product of homologies is scalar, and $S_k \subset V_k$ the closed subset of those preserving a conic. Throughout, $M = M(L_\bullet, O_\bullet)$ denotes the $3k \times 6$ matrix of the linear system in the six unknowns $Q_{11}, Q_{12}, Q_{13}, Q_{22}, Q_{23}, Q_{33}$ whose i-th block of three rows expresses the vanishing of the cross product $QO_i \times L_i^\top$, so that $\ker M$ is the space of invariant forms and $S_k = \{\operatorname{rank} M \leq 5\}$. Both dimensions below are computed at explicit rational points; semicontinuity propagates the values to the component through each point.

Theorem 5.5. *At the configuration of Theorem 5.2 ($k = 5$) and at that of Remark 5.4 ($k = 6$):*

$$\dim V_k = 4k - 8, \qquad \dim(\textit{moduli}) = 4k - 16, \qquad \dim S_k = 2k + 2,$$

and these values hold on a Zariski-open subset of the component through each point.

Proof. Parametrise a configuration by $(L_i, O_i) \in (\mathbb{R}^3)^{2k}$, which is $6k$ coordinates with $2k$ scaling directions acting trivially. The condition that the product be scalar is the vanishing of the traceless part of P, i.e. 8 equations. At each of the two points the Jacobian of those 8 equations has rank exactly 8, computed over $\mathbb{Q}$ by Gaussian elimination; rank being lower semicontinuous, it is 8 on a Zariski-open subset. Hence $\dim V_k = 6k - 2k - 8 = 4k - 8$ there. The infinitesimal stabiliser, the space of $X \in \mathfrak{gl}_3$ with $XO_i \propto O_i$ and $L_i X \propto L_i$ for all i, is computed exactly to be one-dimensional, i.e. spanned by I, so the PGL_3-stabiliser is trivial; being upper semicontinuous, it is trivial generically, and the moduli dimension is $(4k - 8) - 8 = 4k - 16$.

For S_k, fix the conic Q and parametrise by $(n_1, \ldots, n_k)$, which is $2k$ projective parameters. Every reflection then lies in $\mathrm{O}(Q)$ by Lemma 4.3, so the product does too, and the closure condition is at most $\dim \mathrm{O}(Q) = 3$ conditions; at both points the Jacobian in the n_i directions has rank exactly 3 over $\mathbb{Q}$, attaining the bound. As the invariant conic is unique up to scale at both points — the solution space of $L_i \propto (QO_i)^\top$ is one-dimensional, again computed exactly — letting Q range over the 5-dimensional space of conics gives $\dim S_k = 5 + 2k - 3 = 2k + 2$. □

Corollary 5.6. $2k + 2 = 4k - 8$ *if and only if* $k = 5$. *Consequently:*

(i) *for $k = 5$, S_5 has the same dimension as V_5, hence is a union of irreducible components of V_5; in particular the component through the configuration of Theorem 5.2, which preserves a conic, is contained in S_5, so every configuration in* that component *preserves a conic. Whether the conic is nondegenerate at every point of the component is a separate question; see Remark 5.43;*

(ii) *for $k \geq 6$, S_k is a proper closed subset of V_k, so the generic configuration on the component through the point of Remark 5.4 preserves no conic, and therefore comes from no geometry of constant curvature.*

5.5 The threshold at $k = 5$

Corollary 5.6(i) says that one component of V_5 lies in S_5. We now prove a statement that does not depend on irreducibility, valid on an explicit open subset of V_5. The mechanism is to group the five mirrors as $2 + 2 + 1$.

Recall first that the involutions of PGL_3 are exactly the harmonic homologies, and note the following normal form, which also removes the fifth mirror from the problem.

Proposition 5.7. *Let $H_0, \dots, H_3$ be harmonic homologies and $M = H_3H_2H_1H_0$. There is a harmonic homology H_4 with $H_4H_3H_2H_1H_0 = -I$ if and only if $M^2 = I$ and $M \neq I$; then $H_4 = -M^{-1}$ is unique. Consequently* $\dim V_5 = 16 - 4 = 12$*, since* $\dim \mathrm{Inv}^4 = 16$ *and the involutions form a subvariety of codimension* $8 - 4 = 4$ *in* PGL_3*, in agreement with Theorem 5.5.*

Proof. $H_4 := -M^{-1}$ gives the relation, and $\det H_4 = (-1)^3 \det M^{-1} = -1$ since $\det M = 1$. So H_4 is a harmonic homology iff $H_4^2 = I$, i.e. $M^2 = I$; and $M = I$ gives $H_4 = -I$, trivial in PGL_3. □

Lemma 5.8. *Let H_0, H_1 be harmonic homologies and $A = H_1H_0$. Then $H_0AH_0 = A^{-1}$ and A has eigenvalue 1. If moreover A is diagonalisable with eigenvalues $1, \lambda, \lambda^{-1}$ and $\lambda \neq \pm 1$, then in an eigenbasis (e_1, e_2, e_3) the conics preserved by A form the pencil*

$$\mathcal{P}_A = \{\alpha\, x_1^2 + 2\beta\, x_2x_3\},$$

and every member of $\mathcal{P}_A$ is preserved by H_0 and by H_1. Thus $\mathcal{P}_A$ is exactly the set of conics preserved by both.

Proof. $H_0AH_0 = H_0H_1H_0H_0 = H_0H_1 = (H_1H_0)^{-1} = A^{-1}$. Hence A is conjugate to A^{-1}, so its spectrum is stable under $\mu \mapsto \mu^{-1}$; in dimension 3 some eigenvalue is fixed by that involution, hence equals ± 1, and $\det A = 1$ excludes -1, since then the remaining two would be mutually inverse with product $1 \neq -1$. So 1 is an eigenvalue.

In the eigenbasis, $\tau_A(Q) = A^\top QA$ multiplies the entry Q_{ij} by $\lambda_i\lambda_j$, so Q is fixed exactly when $Q_{ij} = 0$ unless $\lambda_i\lambda_j = 1$; for $\lambda \neq \pm 1$ this leaves Q_{11} and Q_{23}, i.e. the stated pencil.

Since H_0 conjugates A to A^{-1}, it maps the μ-eigenspace of A to the μ^{-1}-eigenspace: it fixes e_1 and interchanges e_2, e_3. So in that basis $H_0 = \begin{pmatrix} p & & \\ & & q \\ & r & \end{pmatrix}$, and $H_0^2 = I$ forces $p^2 = 1$ and $qr = 1$. A direct computation then gives $H_0^\top(x_1^2)H_0 = p^2x_1^2 = x_1^2$ and $H_0^\top(x_2x_3)H_0 = qr\, x_2x_3 = x_2x_3$, so τ_{H_0} is the identity on $\mathcal{P}_A$; and $H_1 = AH_0$ then preserves it too. Conversely a conic preserved by both is preserved by A. □

Theorem 5.9. *Let $H_0, \dots, H_4$ be harmonic homologies with $H_4H_3H_2H_1H_0 = -I$, and suppose $A = H_1H_0$ and $B = H_3H_2$ are diagonalisable with eigenvalues $1, \lambda^{\pm 1}$ and $1, \mu^{\pm 1}$ respectively, $\lambda, \mu \neq \pm 1$. Then $H_0, \dots, H_4$ preserve a common* ***nonzero quadratic form***

$$Q = \alpha\, x_1^2 + 2\beta\, x_2x_3, \qquad \alpha = \frac{a}{w_1}, \qquad \beta = \frac{b}{w_3} \quad \text{or} \quad \frac{c}{w_2},$$

in the notation of the proof. If $w_2 = w_3 = 0$, the displayed formula is replaced by $Q = x_2x_3$. Otherwise β is defined using whichever of w_2, w_3 is nonzero; the relation $b\,w_2 = c\,w_3$ makes the two definitions agree when both apply. Nondegeneracy of Q is not read off from these coordinates; under the geometric hypotheses of Lemma 5.11 it follows from that lemma.

Proof. Work in an eigenbasis of A, so $A = \mathrm{diag}(1, \lambda, \lambda^{-1})$ and, by Lemma 5.8, the conics preserved by H_0 and H_1 are the pencil $\mathcal{P}_A = \{\alpha x_1^2 + 2\beta x_2x_3\}$.

Put $N = BA$. By Proposition 5.7, $N^2 = I$; also $\det N = 1$ and $N \neq I$, so N has eigenvalues $(1, -1, -1)$ and can be written $N = 2P - I$ with $P = w\tilde{w}/s$ of rank one, $s = \tilde{w}w \neq 0$. Write $w = (w_1, w_2, w_3)^\top$ and $\tilde{w} = (a, b, c)$.

If $w_2 = w_3 = 0$, then the lower 2×2 block of N is $-I$ and the form $Q = x_2x_3$ is preserved by A, by N, and hence by $B = NA^{-1}$. Since B is regular semisimple, Lemma 5.8 gives preservation by H_2, H_3, and then by H_4. Assume henceforth that $(w_2, w_3) \neq (0, 0)$.

When does B preserve a member of $\mathcal{P}_A$? Since $B = NA^{-1}$ and $A^\top QA = Q$ for $Q \in \mathcal{P}_A$, we have $B^\top QB = (A^{-1})^\top (N^\top QN)A^{-1}$, which equals Q if and only if $N^\top QN = Q$. *The relation forces the conic condition.* By Lemma 5.8 applied to the pair (H_2, H_3), B has eigenvalue 1, i.e. $\det(B - I) = 0$. As $B - I = (N - A)A^{-1}$ and $\det A = 1$, this says $\det(N - A) = 0$. Now $N - A = -D + \frac{2}{s} w\tilde{w}$ with $D = I + A = \operatorname{diag}(2, 1+\lambda, 1+\lambda^{-1})$, so by the rank-one determinant formula

$$\det(N - A) = \det(-D)\Big(1 - \tfrac{2}{s}\,\tilde{w}D^{-1}w\Big) = -\det D \cdot \frac{s - 2\tilde{w}D^{-1}w}{s}.$$

With $D^{-1} = \operatorname{diag}\big(\frac{1}{2}, \frac{1}{1+\lambda}, \frac{\lambda}{1+\lambda}\big)$ the numerator is

$$s - 2\tilde{w}D^{-1}w = bw_2 + cw_3 - \frac{2(bw_2 + \lambda c w_3)}{1+\lambda} = \frac{(\lambda - 1)(bw_2 - cw_3)}{1+\lambda},$$

the terms in aw_1 cancelling. Since $\det D = 2(1+\lambda)^2/\lambda$,

$$\boxed{\det(N - A) = -\frac{2(1+\lambda)(\lambda - 1)\,(b\,w_2 - c\,w_3)}{\lambda\, s}}$$

$$b\,w_2 = c\,w_3. \tag{4}$$

As $\lambda \neq \pm 1$ and $s \neq 0$, $\det(N - A) = 0$ is *equivalent* to (4).

For $Q = \alpha x_1^2 + 2\beta x_2 x_3$ one computes $w^\top Q = (\alpha w_1,\ \beta w_3,\ \beta w_2)$. If $w_1 = 0$, the relation (4) and a direct trace calculation give $\operatorname{tr} B = -1$, contradicting $\mu \neq \pm 1$; hence $w_1 \neq 0$. Thus $\alpha = a/w_1$, while β is determined by the nonzero one of w_2, w_3, and the two remaining components are compatible by (4). (If exactly one of w_2, w_3 vanishes, say $w_2 = 0$, then (4) reads $c\,w_3 = 0$, hence $c = 0$, and the remaining components determine β.)

For the resulting Q, one has $\tilde{w} = \rho w^\top Q$ for some $\rho \neq 0$. Since $s = \tilde{w}w \neq 0$, also $w^\top Q w \neq 0$, and direct substitution gives

$$N = 2\frac{w w^\top Q}{w^\top Q w} - I, \qquad N^\top Q N = Q.$$

This direct computation is valid even when Q is degenerate; no converse form of the polarity lemma is being used.

Hence there is $Q \in \mathcal{P}_A$ preserved by B. By Lemma 5.8 it is preserved by H_0 and H_1; being preserved by B and lying in $\mathcal{P}_B$ — again by Lemma 5.8, the conics preserved by B are exactly those preserved by H_2 and H_3 — it is preserved by H_2 and H_3; and therefore by $H_4 = -(BA)^{-1}$. □

Write $V_5^{\mathrm{reg}} \subset V_5$ for the open subset where A and B are regular semisimple with $\lambda, \mu \neq \pm 1$.

Corollary 5.10. *Every configuration in V_5^{reg} preserves a nonzero invariant quadratic form, so $\operatorname{rank} M \leq 5$ there; since that is a closed condition, it also holds on $\overline{V_5^{\mathrm{reg}}}$. If the configuration is a convex pentagon with pairwise distinct centres and no four centres collinear, Lemma 5.11 makes the form nondegenerate and the billiard has constant curvature.*

5.6 Closing the two gaps

Lemma 5.11 (Nondegeneracy is geometric)**.** *Let $Q \neq 0$ be an invariant form of a configuration with $L_i O_i \neq 0$ for all i, and suppose Q is degenerate, with kernel K. Then for each i* exactly one *of the following holds: $O_i \in K$, or $K \subset L_i$. Consequently:*

(i) *if* $\dim K = 1$*, say* $K = \langle p \rangle$*, then at least two of the centres coincide (both equal to* p*);*
(ii) *if* $\dim K = 2$*, at least four of the centres are collinear.*
Hence if the centres are pairwise distinct and no four of them are collinear, every nonzero invariant form is a ***nondegenerate*** *conic.*

Proof. Invariance gives $QO_i = \mu_i L_i^\top$ for scalars μ_i, possibly zero: indeed $H_i^\top Q H_i = Q$ expands to $L_i^\top v^\top + v L_i = \kappa L_i^\top L_i$ with $v = QO_i$, and the left side has rank 2 unless $v \parallel L_i^\top$, while the right side has rank ≤ 1. If $\mu_i = 0$ then $O_i \in K$. If $\mu_i \neq 0$ then $L_i^\top \in \operatorname{im} Q = K^\perp$, i.e. L_i vanishes on K, that is $K \subset L_i$. The two are exclusive: $O_i \in K \subset L_i$ would give $L_i O_i = 0$.

(i) Let $T = \{i : p \in L_i\}$. If $|T| \geq 4$, then among four of the five sides there are two adjacent pairs, and p would equal two distinct vertices of the polygon. So $|T| \leq 3$ and $|S| = 5 - |T| \geq 2$ where $S = \{i : O_i = p\}$.

(ii) Here K is a line ℓ, and $K \subset L_i$ forces $L_i = \ell$; the sides being distinct, this happens for at most one i. So at least four indices have $O_i \in \ell$. □

Proposition 5.12 (Every component has dimension at least 12)**.** *Every irreducible component of* V_5 *has dimension* ≥ 12*.*

Proof. By Proposition 5.7, $V_5 = \varphi^{-1}(\mathrm{Inv})$ with $\varphi : \mathrm{Inv}^4 \to \mathrm{PGL}_3$ the product map. A harmonic homology is a pair (L, O) with $LO \neq 0$ modulo two scalings, so $\dim \mathrm{Inv} = 6 - 2 = 4$ and Inv has codimension $8 - 4 = 4$ in PGL_3; also Inv^4 is irreducible of dimension 16. Locally Inv is cut out by 4 equations, so every component of the preimage has codimension at most 4 in Inv^4, i.e. dimension at least $16 - 4 = 12$. □

Proposition 5.13 (Where regularity fails)**.** $A = H_1 H_0$ *fails to be regular semisimple only if* $\lambda = -1$ *or* $\lambda = 1$*. One has* $A^2 = I$ *if and only if* H_0 *and* H_1 *commute; among such pairs,* $A = I$ *(so* $\lambda = 1$*) exactly when* $H_1 = H_0$*, and* $\lambda = -1$ *exactly when* $H_1 \neq H_0$*. If* $\lambda = 1$ *with* $A \neq I$ *then* A *is unipotent. The commuting locus has dimension* 6 *inside the* 8*-dimensional* Inv^2*, and the unipotent locus dimension* 7*.*

Proof. The spectrum of A is $\{1, \lambda, \lambda^{-1}\}$ by Lemma 5.8, so regular semisimplicity fails exactly when $\lambda = \pm 1$. Now $A^2 = I$ if and only if $H_1 H_0 = (H_1 H_0)^{-1} = H_0 H_1$, i.e. iff the two commute; and then $A = I$ iff $H_1 = H_0$, which is the case $\lambda = 1$, while $A \neq I$ forces the eigenvalues $(1, -1, -1)$, i.e. $\lambda = -1$. Two commuting involutions of PGL_3 are simultaneously diagonalisable, so the pair is determined by a projective frame modulo the diagonal torus: dimension $8 - 2 = 6$. If $\lambda = 1$ then A is unipotent; the regular unipotent class of PGL_3 has dimension $8 - 2 = 6$, of codimension 1 in the 7-dimensional locus of elements with eigenvalue 1, whose preimage in Inv^2 is all of Inv^2; so the unipotent locus has dimension $8 - 1 = 7$. □

Lemma 5.14 (Factorisation fibres)**.** *Let* X *be a product of two harmonic homologies, and let*

$$\mathcal{F}_X = \{(J_0, J_1) \in \mathrm{Inv}^2 : J_1 J_0 = X\}.$$

Using the determinant-one lift of X*, the following are the maximal dimensions of the fibres:*

Jordan type of X	$\dim \mathcal{F}_X$
I	4
$\mathrm{diag}(1, -1, -1)$	2
$J_2(1) \oplus [1]$	2
$J_3(1)$	1
$\mathrm{diag}(1, \rho, \rho^{-1}),\ \rho \neq \pm 1$	1.

The non-semisimple type with eigenvalues $(1, -1, -1)$ *has fibre dimension* 1*.*

Proof. The condition on J_0 is $J_0X = X^{-1}J_0$, together with $J_0^2 = I$ and $\operatorname{tr} J_0 = 1$; then $J_1 = XJ_0$. In the regular semisimple case the solutions have the form

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & q \\ 0 & q^{-1} & 0 \end{pmatrix}, \qquad q \neq 0,$$

up to projective sign. For $J_3(1)$ they are

$$\begin{pmatrix} 1 & 1-q & q(q-1)/2 \\ 0 & -1 & q \\ 0 & 0 & 1 \end{pmatrix},$$

and for $J_2(1) \oplus [1]$ they lie in the two two-parameter families

$$\begin{pmatrix} -1 & r & s \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & r & 0 \\ 0 & -1 & 0 \\ 0 & s & 1 \end{pmatrix}.$$

For $\operatorname{diag}(1,-1,-1)$ the generic solutions are $\operatorname{diag}(1, J)$ with $J = \left(\begin{smallmatrix} a & b \\ c & -a \end{smallmatrix}\right)$ and $a^2 + bc = 1$, a two-dimensional quadric. The identity has the full four-dimensional space Inv. For the non-semisimple type, take $X = \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{smallmatrix}\right)$. The two one-parameter families

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & r \\ 0 & 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & r \\ 0 & 0 & -1 \end{pmatrix}$$

solve $J_0X = X^{-1}J_0$, $J_0^2 = I$ and $\operatorname{tr} J_0 = 1$; these are all the solutions, so the fibre has dimension 1. These normal forms give the table. □

Proposition 5.15 (Global dimension of the non-regular locus)**.** *Let $Z \subset V_5$ be the locus where A or B fails to be regular semisimple. Then every irreducible component of Z has dimension at most* 11*. Consequently V_5^{reg} is dense in every irreducible component of V_5.*

Proof. We prove the assertion for the branch where A is non-regular; the branch for B is identical after cyclically relabelling the five homologies. Write $N = BA$ for the determinant-one lift of the involution in Proposition 5.7, and put

$$\mathcal{C}_A = \{N \in \mathrm{Inv} : \det(NA^{-1} - I) = 0\}.$$

The determinant condition is necessary because $B = NA^{-1} = H_3H_2$ has eigenvalue 1. The five non-regular Jordan types of A and the corresponding dimensions of the pair (H_0, H_1) are

A	$\dim\{(H_0,H_1)\}$	$\dim \mathcal{C}_A$	$\dim\{(N,H_2,H_3) : H_3H_2 = NA^{-1}\}$
I	4	4	6
$\operatorname{diag}(1,-1,-1)$	6	4	≤ 5
$J_3(1)$	7	3	≤ 4
$J_2(1) \oplus [1]$	6	3	≤ 5
$(1) \oplus J_2(-1)$	7	3	≤ 4

Here is the exact calculation behind the last two columns. Write $N = 2w\tilde{w}/(\tilde{w}w) - I$ and, on the chart $w = (1, u, v)$, write $\tilde{w} = (1 - pu - qv, p, q)$. For $A = J_3(1)$, $\det(N - A)$ is twice

$$2pu^2 - puv - 2pv + 2quv - qv^2 - 2u + v,$$

an irreducible nonzero equation; neither $\operatorname{tr}(NA^{-1}) - 3$ nor $\operatorname{tr}(NA^{-1}) + 1$ contains it as a factor. Thus $\dim \mathcal{C}_A = 3$, and the non-regular sublocus of B has dimension at most 2. For $A = J_2(1) \oplus [1]$, the determinant is $4u(pu+qv-1)$ and $\operatorname{tr}(NA^{-1})+1$ vanishes on it, so every such B has a factorisation fibre of dimension at most 2, giving the bound 5. The other two affine charts give the same conclusions. For the non-semisimple type $A = (1) \oplus J_2(-1)$, the triples $(\det(N-A), \operatorname{tr}(NA^{-1}) - 3, \operatorname{tr}(NA^{-1}) + 1)$ in the three charts are

$$
\begin{aligned}
&(-4pv,\ -2(2pu + pv + 2qv),\ -2(2pu + pv + 2qv - 2)),\\
&\big(4v(pu + qv - 1),\ 2(puv + 2pu + qv^2 - v - 2),\ 2(puv + 2pu + qv^2 - v)\big),\\
&(-4q,\ 2(2pu - q - 2),\ 2(2pu - q)).
\end{aligned}
$$

In each line the determinant is nonzero and neither trace polynomial has a common irreducible factor with it. Hence $\dim \mathcal{C}_A = 3$, its regular part has factorisation fibres of dimension 1, and its non-regular part has dimension at most 2 and fibres of dimension at most 2, giving the bound 4. For $A = \operatorname{diag}(1, -1, -1)$ the determinant vanishes identically, but the non-regular locus of $B = NA$ is a proper closed subset of the four-dimensional Inv; its fibres have dimension at most 2, while the regular part has fibres of dimension 1. The exceptional point $B = I$ has a four-dimensional factorisation fibre, but its inverse image is only the single point $N = A$, so this still gives the bound 5. The case $A = I$ gives $B = N$, a non-trivial involution, and has dimension $4 + 2 = 6$.

The first column follows from the conjugacy-class dimensions and Lemma 5.14: the classes of a regular unipotent, a subregular unipotent, a non-trivial involution, and the non-semisimple $(1) \oplus J_2(-1)$ type have dimensions $6, 4, 4, 6$, respectively. Adding the last column to the second gives at most $10, 11, 11, 11, 11$; hence the whole A-branch has dimension at most 11. The same holds for the B-branch. Since Proposition 5.12 gives dimension at least 12 for every component of V_5, no component is contained in Z, proving density. □

Corollary 5.16 (Density)**.** *Let $Z \subset V_5$ be the locus where A or B fails to be regular semisimple. The regular locus V_5^{reg} is dense in every irreducible component of V_5.*

Proof. This is Proposition 5.15. □

Theorem 5.17. *Let $H_0, \dots, H_4$ be harmonic homologies with $H_4H_3H_2H_1H_0 = -I$ bounding a convex pentagon, with pairwise distinct centres and no four centres collinear. Then the five mirrors preserve a* ***nondegenerate*** *conic, and the billiard is one of constant curvature — in the Cayley–Klein sense, for which see Remark 4.11: the signature of the conic decides the geometry, and only the definite case is a metric billiard on the whole domain.*

Proof. By Proposition 5.15, the configuration lies in $\overline{V_5^{\mathrm{reg}}}$. By Corollary 5.10 and the closedness of $\operatorname{rank} M \le 5$ there is a nonzero invariant form. Lemma 5.11 and the hypotheses on the centres make it nondegenerate. □

5.7 Classification and irreducibility at $k = 5$

Everything above was arranged so as not to need irreducibility of V_5. It is in fact irreducible, and the proof is a fibration of V_5^{reg} over the possible values of A; the same coordinates then show that Proposition 5.1 is *exhaustive*, so $k = 5$ is completely classified. All algebraic varieties in the irreducibility argument are complexified; irreducibility over $\mathbb{C}$ implies irreducibility over $\mathbb{R}$, and does not assert connectedness of the real locus. We use geometrically irreducible generic fibres and explicitly exclude components supported over proper subsets of the base. Equal fibre dimensions alone would not suffice.

Lemma 5.18. Inv *is the open subset* $\{(w,\tilde{w}) \in \mathbb{RP}^2 \times \mathbb{RP}^2 : \tilde{w}w \neq 0\}$ *via* $N = 2w\tilde{w}/(\tilde{w}w) - I$, *irreducible of dimension* 4. *If* A *is regular semisimple with eigenvalues* $1, \lambda^{\pm 1}$ *and* $\lambda \neq \pm 1$, *then*

$$\mathcal{C}_A = \{N \in \mathrm{Inv} : \det(N - A) = 0\}$$

is a nonempty proper ***irreducible*** *hypersurface of* Inv, *of dimension* 3.

Proof. That N satisfies $Nw = w$ and $N = -I$ on $\ker \tilde{w}$, so it is the harmonic homology of centre w and axis $\tilde{w}$ up to the sign that is trivial in PGL_3; every element of Inv has this form and $\tilde{w}w \neq 0$ is $LO \neq 0$. So Inv is open in $\mathbb{RP}^2 \times \mathbb{RP}^2$, irreducible of dimension 4.

In an eigenbasis of A the boxed formula in the proof of Theorem 5.9 reads $\det(N - A) = -2(1+\lambda)(\lambda - 1)(bw_2 - cw_3)/(\lambda s)$ with $\tilde{w} = (a, b, c)$, so with $\lambda \neq \pm 1$ and $s \neq 0$ the equation $\det(N - A) = 0$ is *exactly* $bw_2 = cw_3$, a form of bidegree $(1,1)$ on $\mathbb{RP}^2 \times \mathbb{RP}^2$. Such a form is reducible only if it factors as (a linear form in w)(a linear form in $\tilde{w}$) — the only splitting available in bidegree $(1,1)$ — that is, only if its coefficient matrix has rank 1. That matrix is here $\mathrm{diag}(0, 1, -1)$, of rank 2. So $bw_2 - cw_3$ is irreducible and $\mathcal{C}_A$ is an irreducible hypersurface of dimension 3. It is nonempty ($w = \tilde{w} = e_1$) and proper ($w = \tilde{w} = e_2$ gives $bw_2 - cw_3 = 1$). □

Theorem 5.19 (Irreducibility)**.** V_5 *is irreducible, of dimension* 12.

Proof. By Proposition 5.15, V_5^{reg} is dense in every irreducible component of V_5; hence $V_5 = \overline{V_5^{\mathrm{reg}}}$ and it suffices to prove V_5^{reg} irreducible. Also $\dim V_5^{\mathrm{reg}} \geq 12$, by Proposition 5.12 and that same density.

Let $\pi : V_5^{\mathrm{reg}} \to \mathrm{PGL}_3$, $\pi(H_0, \ldots, H_4) = A = H_1 H_0$, with image $\mathcal{A}$. By Lemma 5.8 every element of $\mathcal{A}$ is regular semisimple with eigenvalues $1, \lambda^{\pm 1}$, $\lambda \neq \pm 1$; let $\mathcal{T}$ be the set of all such. $\mathcal{T}$ is the image of $\mathrm{PGL}_3 \times (\mathbb{A}^1 \setminus \{0, \pm 1\}) \to \mathrm{PGL}_3$, $(g, \lambda) \mapsto g\,\mathrm{diag}(1, \lambda, \lambda^{-1})g^{-1}$, hence irreducible; the fibres are cosets of a maximal torus up to a finite group, so $\dim \mathcal{T} = 8 + 1 - 2 = 7$.

The fibre. Fix $A \in \mathcal{A}$. A point of $\pi^{-1}(A)$ is a triple (H_0, N, H_2) with $H_0 \in \mathcal{F}_A$, $N \in \mathcal{C}_A$ such that $B := NA^{-1}$ is regular semisimple with $\mu \neq \pm 1$, and $H_2 \in \mathcal{F}_B$; then $H_1 = AH_0$, $H_3 = BH_2$ and $H_4 = -N^{-1}$ are determined. Indeed $N = M = H_3H_2H_1H_0$ lies in Inv by Proposition 5.7, which also gives $H_4 = -M^{-1}$; and $B = NA^{-1} = H_3H_2$ has eigenvalue 1 by Lemma 5.8, which is $\det(N - A) = 0$ since $\det A = 1$. By Lemma 5.14 the fibres $\mathcal{F}_A$ and $\mathcal{F}_B$ are, in the regular semisimple case, the punctured line $\{q \neq 0\}$: irreducible of dimension 1. By Lemma 5.18, $\mathcal{C}_A$ is an irreducible 3-fold. To see that the admissible locus is nonempty, work over an algebraic closure and conjugate A to $\mathrm{diag}(1, \lambda, \lambda^{-1})$. The involution $N_0 = \mathrm{diag}(1, -1, -1)$ lies in $\mathcal{C}_A$, and

$$B_0 = N_0 A^{-1} = \mathrm{diag}(1, -\lambda^{-1}, -\lambda)$$

has three distinct eigenvalues because $\lambda \neq \pm 1$. Thus B_0 is regular semisimple. Since regular semisimplicity is an open condition, the admissible N form a nonempty open subset of $\mathcal{C}_A$. The family of factorisations of B is smooth of relative dimension one on this open subset: after an etale base change ordering the three simple eigenvalues and choosing local eigenvectors, the formula of Lemma 5.14 identifies it with the product with $\mathbb{G}_m$. Smoothness descends under this base change. In particular the projection is open, so every component dominates the base; its geometrically irreducible generic fibre then permits only one component. Taking the product with $\mathcal{F}_A$ proves that $\pi^{-1}(A)$ is geometrically irreducible of dimension $1+3+1 = 5$.

No vertical components. Each irreducible component D of V_5^{reg} has dimension at least 12, by Propositions 5.12 and 5.15. Since all fibres of π have dimension 5, $\dim \overline{\pi(D)} \geq \dim D - 5 \geq 7$. The irreducible base $\mathcal{T}$ has dimension 7, hence every D dominates $\mathcal{T}$. The

preceding fibre description remains valid over an algebraic closure of the function field of $\mathcal{T}$, where it is irreducible. Distinct dominating components would give distinct components of that generic fibre. There is therefore exactly one component, of dimension $7 + 5 = 12$. Its closure is V_5. □

Corollary 5.20. $V_5 = S_5$. *Every configuration of five harmonic homologies with scalar product preserves a nonzero quadratic form — not merely those of an open subset.*

Proof. Corollary 5.10 gives $V_5^{\mathrm{reg}} \subseteq S_5$, and $S_5 = \{\operatorname{rank} M \leq 5\}$ is closed in V_5, so $V_5 = \overline{V_5^{\mathrm{reg}}} \subseteq S_5 \subseteq V_5$. □

This supersedes Corollary 5.6(i), which only placed S_5 among the components of V_5.

Theorem 5.21 (The construction is exhaustive)**.** *Let Q be nondegenerate, let $R_i = I - 2n_i n_i^\top Q/(n_i^\top Q n_i)$ be Q-reflections with $R_4R_3R_2R_1R_0 = -I$ and $R_3 \neq R_4$, and put $B = -R_2R_1R_0$. Then $B \neq I$ and there is $w \neq 0$ with $Bw = w$ and $n_4^\top Q w = 0$; consequently $R_3 = BR_4$. So every such tuple arises from the construction of Proposition 5.1, performed with the form Q.*

Proof. Since $(-X)^{-1} = -X^{-1}$, the relation gives $R_4R_3 = -(R_2R_1R_0)^{-1} = B^{-1}$, hence $R_3R_4 = B$. If $B = I$ then $R_3 = R_4$, excluded. Apply Lemma 5.8 to the pair (R_4, R_3), whose product is $R_3R_4 = B$: it gives $R_4BR_4 = B^{-1}$ and that B has eigenvalue 1. So R_4 preserves $E = \ker(B - I) \neq 0$ and acts on it as an involution, whence E is spanned by ±1-eigenvectors of R_4. If all of them had eigenvalue -1 then E would lie in the (-1)-eigenline $\langle n_4 \rangle$, so $E = \langle n_4 \rangle$ and $R_4 w = -w$ for w spanning E; then $Bw = w$ reads $R_3R_4w = w$, i.e. $R_3 w = -w$, so $n_3 \parallel w \parallel n_4$ and $R_3 = R_4$, excluded. Hence some $w \in E \setminus \{0\}$ has $R_4 w = w$, i.e. $n_4^\top Q w = 0$; and $R_3 = R_3R_4R_4 = BR_4$. □

Remark 5.22. Proposition 5.1 was stated for $Q = x^2 + y^2 + z^2$. For an indefinite Q the same construction works, with one hypothesis added: w must be non-isotropic. The step "a reflection whose mirror contains the axis conjugates the rotation to its inverse" is the statement that a reflection of the 2-dimensional space $w^{\perp_Q}$ inverts its rotation group, and that needs $w^{\perp_Q}$ nondegenerate. Both signatures occur among genuine billiards, by Proposition 4.10.

Remark 5.23 (What is classical here, and what is not). Worth saying plainly, because it is the first thing a reader will suspect. In Theorem 4.6 the eigenvalues of W are -1 and $e^{\pm i\theta}$ with θ a *rational* multiple of π, so W has finite order. For the spherical member the mirror group is therefore finite, and it can be identified: at $n = 5$, $j = 2$ the Gram matrix of the five normals in the metric of the invariant conic is circulant with

$$g_{i,i} = 1, \qquad g_{i,i\pm1} = \tfrac{1}{2} = \cos\tfrac{\pi}{3}, \qquad g_{i,i\pm2} = \tfrac{1-\sqrt{5}}{4} = \cos\tfrac{3\pi}{5},$$

and the group generated has order 60 in PGL_3, i.e. 120 in $\mathrm{O}(3)$: it is the icosahedral group H_3, and the five mirrors are five of its fifteen. So the spherical member is a spherical kaleidoscope seen in $\mathbb{RP}^2$, and that is also where its $\sqrt{5}$ comes from. What is *not* classical about it is the reason it exists at all for odd n: that $-I$ is trivial in PGL_3, which is what turns [4, Prop. 1.5] from a wall into a $\mathbb{Z}/2$ obstruction one can pass.

Neither of these applies to the configuration of Theorem 5.2. Its normals are rational, so each $\operatorname{tr}(R_aR_b)$ is rational, and by Niven's theorem the only rational values of $\cos(q\pi)$ with q rational are $0, \pm\frac{1}{2}, \pm1$; eight of the ten pairs fall outside that list, so those products have infinite order and the mirror group is infinite. It is not a kaleidoscope, which is what makes it, and not the symmetric pentagon, the answer to the second question of Problem 1.

5.8 Development: what the relation does to the metric billiard

When the invariant conic is nondegenerate the billiard is a metric one, and the relation $\prod H_i = cI$ then says something sharp about the *lengths* of the orbits. The tool is the classical development of a billiard trajectory, stated here with care, because the naive version — "the unfolded arc runs from p_0 to $g(p_0)$" — is not what is true, and the difference is the whole point.

Lemma 5.24 (Development)**.** *Let M be a Riemannian surface of constant curvature, $\Omega \subset M$ a domain bounded by totally geodesic walls, and let $\gamma : [t_0, t_k] \to \overline{\Omega}$ be a billiard trajectory with bounces at $t_1 < \cdots < t_k$ off the walls $a_1, \dots, a_k$, with reflections $H_{a_j} \in \mathrm{Isom}(M)$. Put $g_0 = \mathrm{id}$, $g_j = H_{a_1} \cdots H_{a_j}$, and define $\sigma(t) = g_j(\gamma(t))$ for $t \in [t_j, t_{j+1}]$. Then σ is a **geodesic** of M of the same length as γ. If moreover γ is a k-periodic orbit, then*

$$\sigma(t_k) = g_k\,\sigma(t_0), \qquad \sigma'(t_k) = g_k\,\sigma'(t_0),$$

*that is, g_k **translates the geodesic σ along itself** by the length of the orbit.*

Proof. Continuity at t_j: the left value is $g_{j-1}\gamma(t_j)$, the right value is $g_j\gamma(t_j) = g_{j-1}H_{a_j}\gamma(t_j) = g_{j-1}\gamma(t_j)$, because $\gamma(t_j)$ lies on the wall a_j, which H_{a_j} fixes pointwise. Continuity of the velocity: the reflection law is $\gamma'(t_j^+) = H_{a_j}\gamma'(t_j^-)$, so $\sigma'(t_j^+) = g_{j-1}H_{a_j}\gamma'(t_j^+) = g_{j-1}H_{a_j}^2\gamma'(t_j^-) = g_{j-1}\gamma'(t_j^-) = \sigma'(t_j^-)$. Each piece is the isometric image of a geodesic, so σ is a C^1 concatenation of geodesics, hence a geodesic; and isometries preserve length.

For a k-periodic orbit, $\gamma(t_k) = \gamma(t_0) =: p_0$ and $\gamma'(t_k^+) = \gamma'(t_0^+)$, with p_0 on the wall a_k. Hence $\sigma(t_k) = g_{k-1}\gamma(t_k) = g_{k-1}H_{a_k}p_0 = g_kp_0 = g_k\sigma(t_0)$; and $\gamma'(t_k^-) = H_{a_k}\gamma'(t_k^+) = H_{a_k}\gamma'(t_0^+)$ gives $\sigma'(t_k) = g_{k-1}H_{a_k}\gamma'(t_0^+) = g_k\sigma'(t_0)$. □

The two constant-curvature cases now behave in *opposite* ways, and for a single reason: what the relation $\prod H_i = cI$, which is the identity of PGL_3, does to the metric.

Theorem 5.25 (What an open family requires)**.** *Let M be a complete simply connected surface of constant curvature and let $\Omega \subset M$ be a domain bounded by finitely many* totally geodesic *walls. Suppose there is an open set of k-periodic orbits, all with the same itinerary $a_1, \dots, a_k$. Then the isometry $g = H_{a_1} \cdots H_{a_k}$ **translates a two-parameter family of geodesics of M along themselves, all by the same positive length**.*

Proof. The itinerary is fixed, so g is one isometry, the same for every orbit of the family. By Lemma 5.24, each orbit develops to a geodesic translated along itself by g, by the length of that orbit; the space of orbits is 2-dimensional, with distinct orbits developing to distinct geodesics. Restrict to a connected component of the family. Let x_i be its successive impact points, and let v_i^- and v_i^+ be the incoming and outgoing unit tangent vectors at x_i. The first-variation formula for total orbit length is

$$dL = \sum_i \langle v_i^- - v_i^+, \dot{x}_i \rangle.$$

Here $\dot{x}_i$ is tangent to wall a_i, while the reflection law makes $v_i^- - v_i^+$ normal to that wall. Every summand vanishes, so $dL = 0$ on the family and the orbit length is constant on the component. Thus g translates the developed geodesics by that same positive length. □

Which isometries can do that is then a question with a three-line answer in each of the three geometries, and the answers are completely different.

Theorem 5.26 (Only the sphere)**.** *In Theorem 5.25:*

(i) $M = \mathbb{H}^2$: ***impossible.*** *A hyperbolic isometry translates only its axis, and elliptic, parabolic and identity isometries translate no geodesic by a positive length. So a geodesic-walled billiard in the hyperbolic plane is never k-reflective, for any k.*

(ii) $M = \mathbb{E}^2$: ***impossible.*** *A translation by $v \neq 0$ translates exactly the geodesics parallel to v, a* one-*parameter family; a glide reflection translates only its axis; rotations and reflections translate none. So a flat-mirror billiard in the Euclidean plane is never k-reflective either.*

(iii) $M = S^2$: ***possible, and only one way.*** *The isometries translating* every *great circle along itself are exactly $\pm I$, with translation lengths π and 2π; every other isometry translates at most one great circle. So $g = \pm I$, and every orbit of the family has length congruent to π or to 0 modulo 2π; see Lemma 5.35 for when the first value is attained.*

Proof. (i) and (ii) are the classification of isometries of $\mathbb{H}^2$ and $\mathbb{E}^2$ together with the observation that a 2-parameter family of geodesics cannot fit inside a 1-parameter one.

(iii) $-I$ maps every great circle to itself and shifts it by π; the identity shifts it by 2π. Conversely a rotation about an axis a translates only the great circle polar to a, and a rotation-reflection other than $-I$ translates at most one. Alternatively, invariance of a great circle is a polynomial condition on its normal. If it holds on an open set, it holds on every normal; a linear map preserving every normal line is scalar. Orthogonality then gives $\pm I$. □

Remark 5.27 (The one fact behind everything). Theorem 5.26 says that among the three constant-curvature geometries the sphere is the only home for k-reflective billiards, and identifies the mechanism exactly: the relation must be $\pm I$ on S^2. Both signs are trivial in PGL_3, whereas $-I$ is the nontrivial antipodal isometry and $+I$ is the identity isometry. That single sentence contains all of the following, which looked like separate phenomena:

(a) odd k exists at all, because $-I$ is trivial in PGL_3: the obstruction of [4, Prop. 1.5] is a $\mathbb{Z}/2$ and not a wall (Theorem 4.6);

(b) the length of every orbit of the family is an odd multiple of π, because that is how far $-I$ moves a great circle (Theorem 5.33), and it is π itself whenever the domain satisfies both the hemisphere and total-chord bound of Lemma 5.35;

(c) nothing exists in the hyperbolic plane, because there the identity of PGL_3 is the identity isometry and moves nothing;

(d) nothing exists for flat mirrors in the Euclidean plane either, which is why Fierobe's Euclidean examples *must* use a transversal field that is not the normal — they are projective billiards and not metric ones;

(e) and in $\mathbb{RP}^n$ with n odd the whole thing collapses to even k, because every real scalar matrix has positive determinant in even vector rank, whereas an odd number of reflections has negative determinant.

So the subject exists because the double cover $S^2 \to \mathbb{RP}^2$ is nontrivial, and the projective setting is not a convenience: it is where the phenomenon lives.

Remark 5.28 (Attribution). Part (ii) is classical — it is the standard unfolding argument for polygonal billiards. Part (i) is elementary and may well be folklore; what is in the literature is the stronger and harder statement for *smooth* boundaries at $k = 3$, proved by Jacobi fields in [3], together with the classification of the 3-reflective spherical billiards as those bounded by three orthogonal great circles. We claim novelty for neither (i) nor (ii); the content here is the trichotomy of Theorem 5.26 and the identification, in Remark 5.27, of $\pm I$ as the whole mechanism.

5.9 Global periodicity of an analytic ambient

The constant-curvature classification above has the following consequence for general analytic ambients. A common refocusing time does not by itself identify the first conjugate locus, so no general roundness conclusion is included here.

Theorem 5.29 (Global periodicity of the ambient)**.** *Let (M^n, g) be connected, complete and real analytic, with $n \geq 2$. Let the walls of a billiard domain be totally geodesic hypersurfaces whose reflections extend to global isometries. Suppose a nonempty open family of regular k-periodic orbits has a common itinerary. On a connected sufficiently small part of that family the length is a constant $\ell > 0$. Let G be the composition of wall reflections in the unfolding convention. Then $dG = \Phi_\ell$ on SM. Moreover:*

(i) $\exp_p(\ell v) = G(p)$ *for every $p \in M$ and unit $v \in T_pM$;*
(ii) $\Phi_{2\ell} = \mathrm{id}$ *and* $G^2 = \mathrm{id}$*;*
(iii) M *is compact and* $\operatorname{diam} M \leq \ell$*;*
(iv) *if* $G = \mathrm{id}$ *then* $\Phi_\ell = \mathrm{id}$*; otherwise G has no fixed point.*

Proof. For a smooth variation of a closed billiard polygon, the first variation of its total length is the sum over vertices of $\langle u_i^- - u_i^+, \delta p_i \rangle$, where u_i^- and u_i^+ are the incoming and outgoing unit velocities. Reflection makes their difference normal to the wall and δp_i tangent to it. Thus the derivative is zero. Smooth dependence of the regular geodesic segments gives a constant length ℓ on a connected local family; no minimizing assumption on the segments is needed for this first variation.

Unfolding gives $dG(v) = \Phi_\ell(v)$ for the initial vectors of the family. A local flow box extending the transverse impact section supplies a nonempty open subset of SM with the same identity. Both maps are analytic and SM is connected for $n \geq 2$, so the identity holds everywhere. Projection to M proves (i).

Let $J(v) = -v$ be velocity reversal. It commutes with dG and satisfies $J\Phi_t J = \Phi_{-t}$. Therefore $\Phi_\ell = dG = JdGJ = \Phi_{-\ell}$, so $\Phi_{2\ell} = \mathrm{id}$. Also $d(G^2) = \mathrm{id}$, and projection gives $G^2 = \mathrm{id}$.

Differentiating (i) in the unit initial direction produces nonzero Jacobi fields vanishing at times 0 and ℓ. Hence every geodesic from p has a conjugate point by time ℓ and cannot minimize beyond that time. Completeness and Hopf–Rinow give $\operatorname{diam} M \leq \ell$ and compactness.

Finally suppose $G(p) = p$. The orthogonal involution dG_p has eigenvalues ± 1. A unit -1 eigenvector would yield $\gamma_v(\ell) = p$ and $\dot\gamma_v(\ell) = -v$. The geodesics $t \mapsto \gamma_v(\ell - t)$ and $t \mapsto \gamma_v(t)$ then have the same initial data, so are equal; differentiating at $\ell/2$ contradicts unit speed. Thus $dG_p = I$, and an isometry fixing a point with identity derivative is the identity on a connected complete manifold. This proves (iv). □

Remark 5.30 (Scope of the conclusion)*.* For the round sphere the two possibilities are the identity and antipodal map; on real projective space the latter induces the identity. These are examples of the theorem, not a classification of all its analytic ambients. To invoke Wiedersehen rigidity one would additionally need to identify the first conjugate locus, a step not proved here. The direct constant-curvature result of Theorem 5.26 remains independent of this issue.

Remark 5.31 (Three branches, and what projectivity buys)*.* Theorem 5.29 is metric: it needs the walls to be totally geodesic for some metric. A projective billiard need not be metric, and comparing which of the examples in these notes escape gives a trichotomy, all three cases of which occur here:

(1) *definite invariant conic*: the associated Cayley–Klein metric is the round metric in projective coordinates, and its spherical lift satisfies the theorem. This is the symmetric pentagon, Theorem 5.2;
(2) *indefinite invariant conic*: pseudo-Riemannian, and the theorem does not apply. This is the branch $\lambda < 0$ of Theorem 4.6, of signature $(2,1)$, and it is where all but one member of the family lives;
(3) *no invariant conic at all*: nothing to apply. This is Fierobe's Euclidean family and, by Corollary 5.6(ii), the generic configuration once $k \geq 6$.

In particular Remark 5.27(d) becomes a theorem rather than an observation: Fierobe's Euclidean examples *must* use a transversal field other than the normal, because with the normal they would be metric billiards in the plane with geodesic walls, which Theorem 5.29 forbids. And branch (2) is not an exception to the mechanism but the same one: the geodesics of a Cayley–Klein geometry whose plane section meets the quadric in an ellipse are closed, of a common projective length, with $-I$ acting as the half-turn, and that holds whatever the signature.

Remark 5.32 (Why odd k was the hard case). Taking determinants, $\det G = (-1)^k$ and $\det(-I_{n+1}) = (-1)^{n+1}$, so in the metric branch $G = -I$ forces $k \equiv n+1 \pmod 2$ and $G = +I$ forces k even. For $n = 2$, within the definite spherical branch, this says k odd $\iff G = -I \iff L \equiv \pi \pmod{2\pi}$, the residue and not the value being what the relation determines (Theorem 5.33). So the odd case is not one case among several: it is exactly the rigid one, whereas even k admits $G = +I$ and with it the Euclidean escape of branch (3), which is where the known examples were. And at $k = 5$ there is no escape by branch (3) under the centre hypotheses of Theorem 5.17, which supply a nondegenerate invariant conic; consistently, the family of Theorem 4.6 has exactly one spherical member for each n and everything else of signature $(2,1)$.

Theorem 5.33 (Spherical: the length is determined modulo 2π)**.** *Let a k-reflective projective billiard preserve a positive definite conic Q, so that by Proposition 2.2 it is the metric billiard of the round metric of Q with totally geodesic mirrors. If $H_{k-1}\cdots H_0 = -I$ then every orbit of the k-periodic family has length $L \equiv \pi \pmod{2\pi}$; if $+I$, $L \equiv 0 \pmod{2\pi}$.*

Proof. Theorem 5.26(iii), or directly: by Lemma 5.24 the developed curve is a great-circle arc translated along itself by $g = \pm I$. A translation of a closed great circle is determined only modulo its length 2π. □

Remark 5.34 (A correction, and where it came from). An earlier version of this statement asserted $L = \pi$ outright. That is not what the development gives: nothing in $g = -I$ distinguishes π from 3π. To conclude $L = \pi$ one must separately show that the developed arcs make only the short half-turn; that geometric estimate is additional to the scalar monodromy relation.

The short branch is recovered by a hypothesis that costs nothing in practice.

Lemma 5.35 (Short branch)**.** *Suppose in addition that the spherical domain lies in an open hemisphere. Then every arc of the orbit is shorter than π, and since $t \mapsto t/(2\sin(t/2))$ increases on $(0,\pi]$ with value $\pi/2$ at the endpoint,*

$$L \;\leq\; \frac{\pi}{2}\sum_j \big|\hat{p}_{j+1} - \hat{p}_j\big|,$$

the sum being of Euclidean chords between the unit vectors of consecutive impacts. Hence if the chords sum to less than 4 then $L < 2\pi$, and with Theorem 5.33 this forces $L = \pi$.

Proof. A convex subset of an open hemisphere has diameter $< \pi$, so each arc is minor and the chord determines it. The stated inequality is $t \le (\pi/2) \cdot 2\sin(t/2)$ on $(0, \pi]$, which is the monotonicity just quoted. Summing and using $\frac{\pi}{2} \cdot 4 = 2\pi$ gives the conclusion. □

Remark 5.36. The point of trading arcs for chords is arithmetic, not geometric: an arc is a transcendental function of rational data, whereas the square of a chord between unit vectors is $2 - 2\langle u, v\rangle$ and is algebraic over $\mathbb{Q}$. So the hypothesis of Lemma 5.35 can be certified exactly, which the value of the arc cannot.

Remark 5.37 (A sharper budget). The constant $\pi/2$ is the worst case and is attained only when an arc equals π. Since $t \mapsto t/(2\sin(t/2))$ increases, it suffices to bound the largest arc: if every arc is at most $\pi/2$ — equivalently, if consecutive impacts have positive inner product, which is a rational condition — then the constant improves to $(\pi/2)/(2\sin(\pi/4)) = \pi/(2\sqrt{2})$ and the hypothesis becomes

$$\sum_j \left|\hat{p}_{j+1} - \hat{p}_j\right| \;<\; 4\sqrt{2} \;=\; 5.656\ldots$$

instead of 4. This matters when one wants to *iterate* a construction: each composition adds chords, the crude budget runs out quickly, and the sharpened one is larger by a factor $\sqrt{2}$ at no cost, since the extra hypothesis is checked by the same rational data. The point came up in checking a composition that already spent 3.081 of the crude budget of 4 after a single step.

Corollary 5.38. *Every orbit of the family of Theorem 5.2 has length exactly π.*

Proof. The pentagon lies in the open hemisphere $z > 0$ of the chart, so Lemma 5.35 applies. The five chords of the witness orbit are bounded above, by exact rational bracketing of the square roots involved, by numbers summing to less than $2.972 < 4$. □

Remark 5.39. Before 9 August 2026 this corollary was asserted as a consequence of Theorem 5.33 alone, and the supporting decimal arc lengths came from a floating-point computation that certifies nothing. The statement was right and the proof was not. It is now checked over $\mathbb{Q}$ by `verificadores/longitud_pi.py`.

Remark 5.40 (Why this is the spectrally interesting example). Ivrii's conjecture, which opened these notes, matters because the second term of the Weyl asymptotics needs the periodic set to have measure zero. A k-reflective billiard with definite conic is a genuine spherical billiard whose periodic set has positive measure, so it is a place where that mechanism can be examined rather than assumed — and by Remark 5.23 the two examples behave completely differently. The symmetric pentagon is a kaleidoscope: it unfolds to a finite group, *every* orbit is periodic, and its spectrum is explicit in spherical harmonics. The configuration of Theorem 5.2 has infinite mirror group, so it does not unfold to anything finite, yet it still carries a two-parameter family of 5-periodic orbits, all of length exactly π by Corollary 5.38. In the trace formula a d-parameter family of periodic orbits of common length contributes with amplitude $\sim k^{d/2}$ against k^0 for an isolated unstable orbit, so the peak of the length spectrum at $L = \pi$ should grow like k. That is a falsifiable statement about an explicit rational configuration, and the exponent is the observable.

Corollary 5.41 (Classification at $k = 5$)**.** *Every convex 5-reflective projective billiard — convex pentagon, pairwise distinct centres, no four centres collinear — is obtained thus: take a nondegenerate Q and n_0, n_1, n_2 with $n_i^\top Q n_i \neq 0$; put $B = -R_2R_1R_0$ and take $w \neq 0$ in $\ker(B - I)$; take n_4 with $n_4^\top Q w = 0$ and $n_4^\top Q n_4 \neq 0$; set $R_3 = BR_4$. The parameter space is irreducible of dimension $2 + 2 + 2 + 1 = 7$ and the moduli space has dimension $7 - \dim \mathrm{O}(Q) = 4 = 4k - 16$, in agreement with Theorem 5.5. In particular the moduli space of convex 5-reflective projective billiards is irreducible and unirational.*

Proof. By Theorem 5.17 there is a nondegenerate invariant conic Q. The first paragraph of the proof of Lemma 5.11 gives $QO_i = \mu_i L_i^\top$, and $\mu_i \neq 0$ because Q is nondegenerate and $O_i \neq 0$; so each mirror is the polar of its centre and H_i is the Q-reflection with normal $n_i = O_i$. The centres being pairwise distinct gives $R_3 \neq R_4$, so Theorem 5.21 applies and the tuple is in the image. The parameter space is an open subset of $(\mathbb{RP}^2)^3 \times \mathbb{RP}^1$, irreducible of dimension 7; the generic PGL_3-stabiliser is trivial by Theorem 5.5, and the subgroup preserving Q is $\mathrm{O}(Q)$, of dimension 3. □

Remark 5.42 (Scope, stated once and plainly). Proposition 5.15 is an exact algebraic dimension argument: it enumerates the five Jordan types of a non-regular A — which by Lemma 5.8 exhaust the possibilities, since A has determinant one and eigenvalue one, so a spectrum $\{1, \lambda, \lambda^{-1}\}$ that is diagonalisable whenever $\lambda \neq \pm 1$ — and bounds each stratum by a conjugacy-class dimension plus a factorisation fibre. It does not rely on the numerical Newton and SVD experiment of `densidad_global.py`, which is retained only as a consistency check and decides nothing. Thus the regular locus is dense in every component of V_5, and Theorem 5.17 is a global statement under its stated geometric hypotheses on the centres. Those hypotheses are imposed explicitly; convexity alone is not used as a substitute for them.

Remark 5.43 (Scope of the classification). Proposition 5.15 and Theorem 5.19 establish the algebraic density and irreducibility assertions. The invariant form is nonzero on all of V_5 by Corollary 5.20; its nondegeneracy in the convex classification uses the distinct-centre and non-collinearity hypotheses of Theorem 5.17. No classification of arbitrary transversal fields, or roundness theorem for every analytic Riemannian ambient, is asserted.

Remark 5.44. The phenomenon is sharp, and the grouping $2 + 2 + 1$ is what makes it work. A single harmonic homology preserves a 4-dimensional space of conics, two of them a 2-dimensional one, and *three generic ones preserve none*; so a common conic for five mirrors is not dimensional slack but a consequence of the relation. The same argument does not apply for $k = 6$, where the natural grouping is $2 + 2 + 2$ and one would need three pencils to meet, consistently with Corollary 5.6(ii).

Reproducibility and computational disclosure

The proofs of the structural theorems are independent of numerical computation. For the explicit rational examples, the supplementary scripts check matrix identities, ranks and strict inequalities in exact arithmetic. Research and manuscript development for all three papers used AI assistance from early July through late August 2026 (approximate). The models used were OpenAI's GPT-6 Astra, GPT-5.6 Sol, and GPT-5.6 Luna, and Anthropic's Fable 5 and Opus 5, accessed through ChatGPT Work, ChatGPT Codex, and Claude Code. No API was used. The assistance supported research, manuscript drafting, development and checking of verification scripts, and revision. Jorge Lucas González reviewed the manuscripts and assumes full responsibility; no AI system is an author.

Conclusion

The paper settles the existence of primitive open families in every period starting at three and gives the complete five-reflective algebraic and convex classification. The metric result identifies the spherical realization and excludes the apparent Klein-disc alternative. The exact certificates support the displayed rational examples, while the structural conclusions do not rely on numerical computation.

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References

[1] A. L. Besse, *Manifolds all of whose geodesics are closed*, Ergebnisse der Mathematik und ihrer Grenzgebiete 93, Springer, 1978.

[2] M. Bialy, C. Fierobe, A. Glutsyuk, M. Levi, A. Plakhov and S. Tabachnikov, *Open Problems on Billiards and Geometric Optics*, Arnold Math. J. **8**(3–4) (2022), 411–422, doi:10.1007/s40598-022-00198-y.

[3] V. Blumen, K. Y. Kim, J. Nance and V. Zharnitsky, *Three-period orbits in billiards on the surfaces of constant curvature*, Int. Math. Res. Not. (2012), 5014–5024, doi:10.1093/imrn/rnr228.

[4] C. Fierobe, *Examples of projective billiards with open sets of periodic orbits*, Discrete Contin. Dyn. Syst. **44** (2024), 3287–3301, doi:10.3934/dcds.2024059.

[5] C. Fierobe, *On projective billiards with open subsets of triangular orbits*, Israel J. Math. **267** (2025), 501–548, doi:10.1007/s11856-024-02712-z; arXiv:2005.02012.

[6] L. W. Green, *Auf Wiedersehensflächen*, Ann. of Math. (2) **78** (1963), 289–299.

[7] V. Ya. Ivrii, *Second term of the spectral asymptotic expansion of the Laplace–Beltrami operator on manifolds with boundary*, Funct. Anal. Appl. **14** (1980), 98–106, doi:10.1007/BF01086550.

[8] S. Tabachnikov, *Introducing projective billiards*, Ergodic Theory Dynam. Systems **17** (1997), 957–976, doi:10.1017/S0143385797086239.